Public Policy 2026-2027

Lecture 8: Fairness (Continued) & Government Failures

Thursday, October 30, 2026
Stockholm School of Economics in Riga

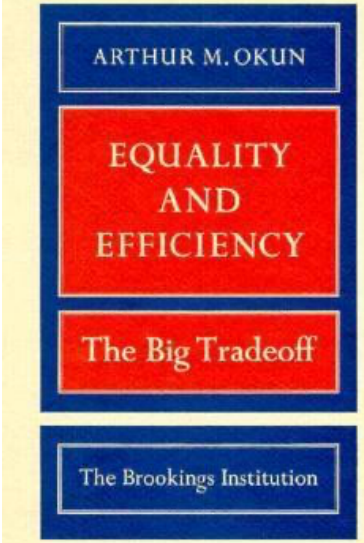
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Arthur Okun's "Leaky Bucket"

At what rate should society trade off equality against efficiency?

- Imagine the government taking money from wealthy citizens through taxes and transferring it to poor citizens
- Inequality-averse people have a preference for more income redistribution. The trouble: transferring €1 from a rich person to a poor person resembles a "leaky bucket":
  - Administrative costs of the transfer instrument.
  - DWL from taxation.
  - Moral hazard: DWL from reduced labor supply.
- At what level of leakage are we no longer willing to undertake this transfer?





1975

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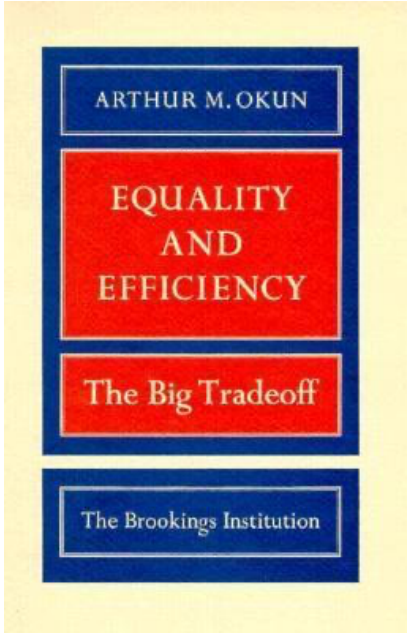
Assuming an isoelastic social valuation of money:

$$(1-L)\,y_P^{-\eta} = y_R^{-\eta}$$

$$L^* = 1 - \left(\frac{y_P}{y_R}\right)^{\eta}$$

y_i = income

η = inequality aversion parameter (= the elasticity of the social marginal utility of income)



1975

Society's Inequality Aversion Parameter

Elasticity of the social marginal utility of income

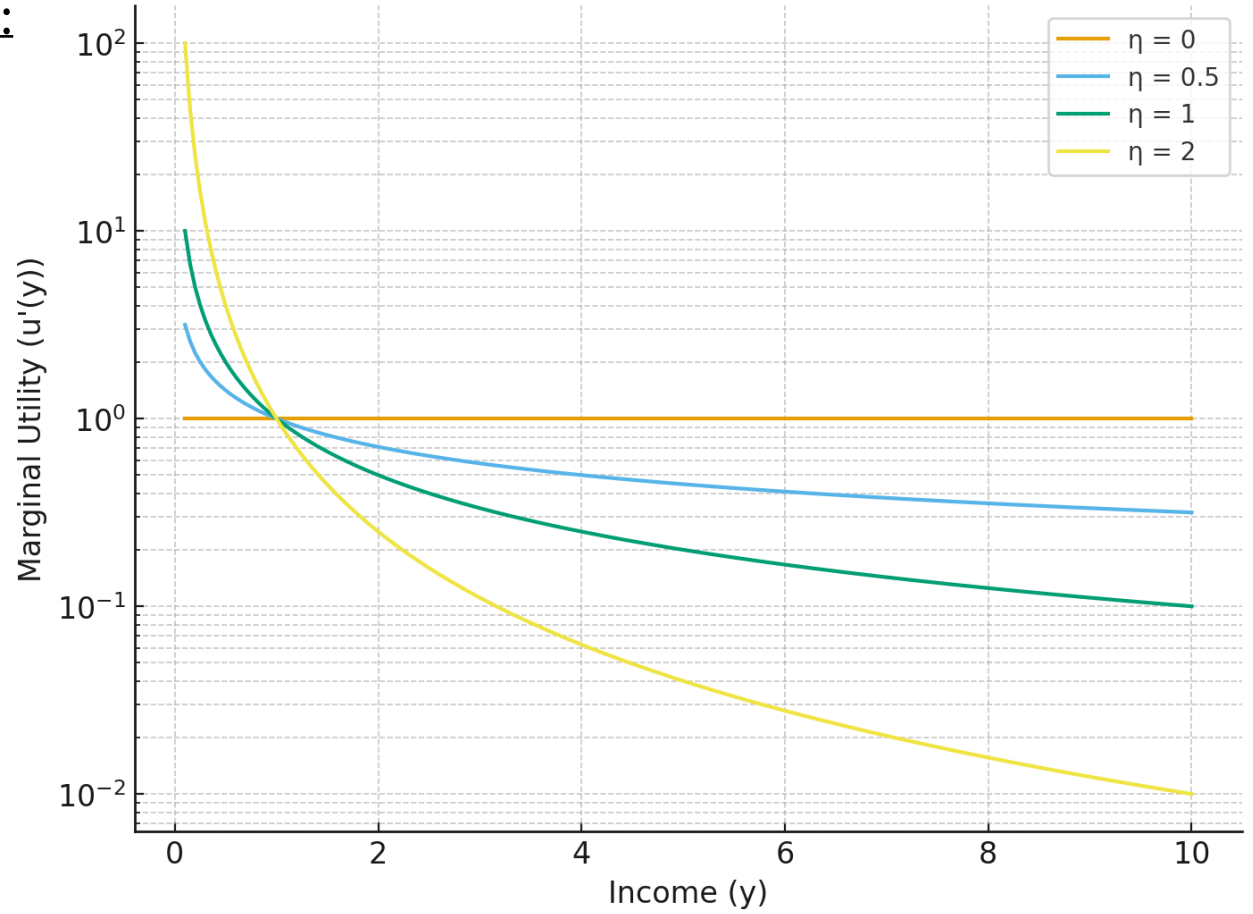
Isoelastic social welfare functions (generic form):

$$W = \sum_i \frac{y_i^{1-\eta}}{1-\eta}$$

for $\eta \neq 1$, and

$$W = \sum_i \ln(y_i)$$

for $\eta = 1$.

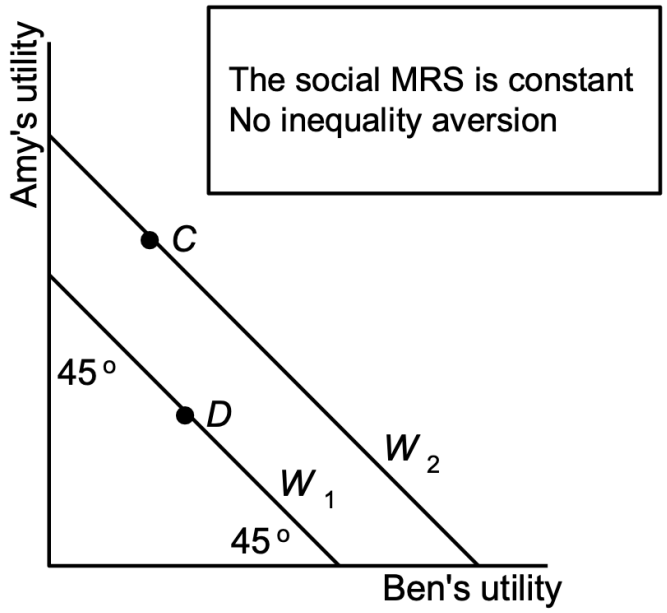


The higher the value of η, the faster the rate at which marginal social utility — **the marginal social value of an extra euro** — falls!

Three Types of Social Indifference Curves

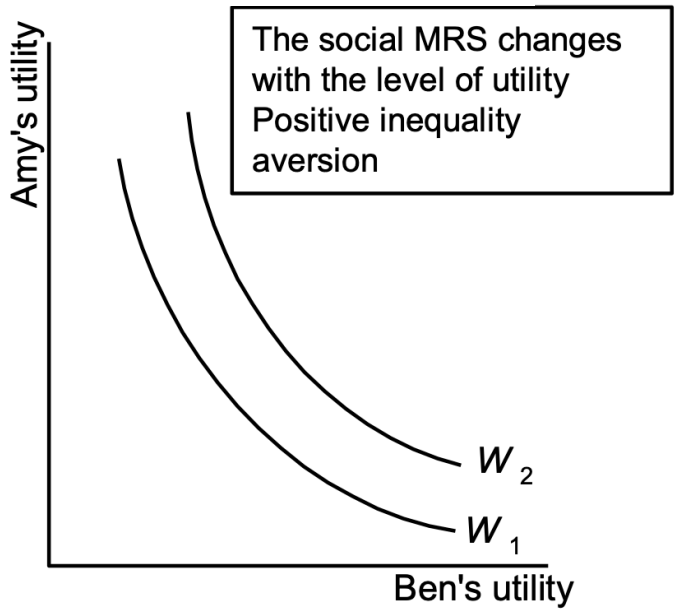
Social utility trade-offs: Bentham, Atksinson/ prioritarian, and Rawls

$$W = \sum_i y_i$$



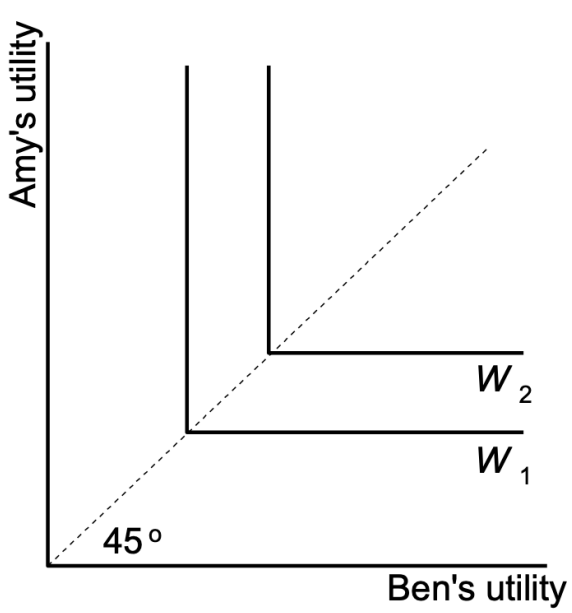
(a) Additive utilitarian SWF

$$W = \sum_i \frac{y_i^{1-\eta}}{1-\eta}$$



(b) Weighted additive utilities SWF

$$W = \min_i y_i$$



(c) Rawlsian SWF

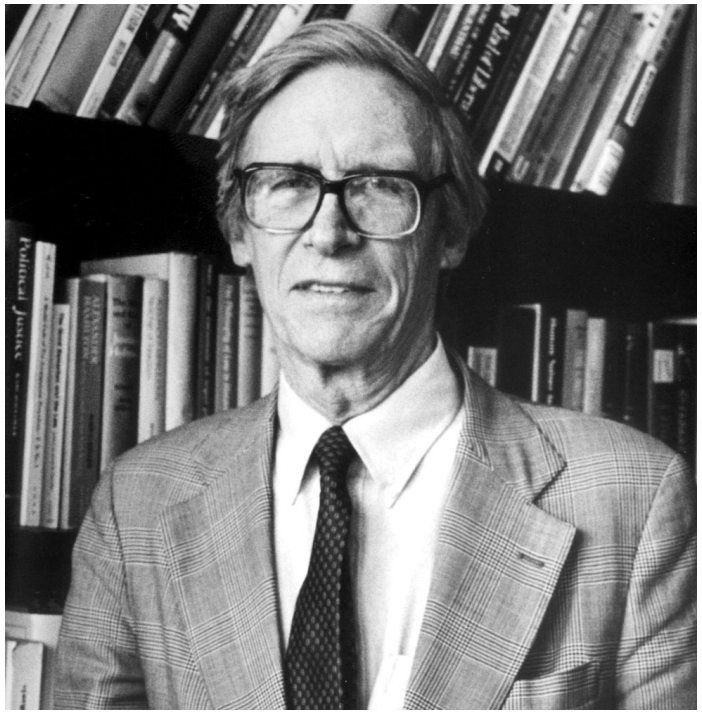
η = 0: complete inequality neutrality

η > 0: some inequality aversion

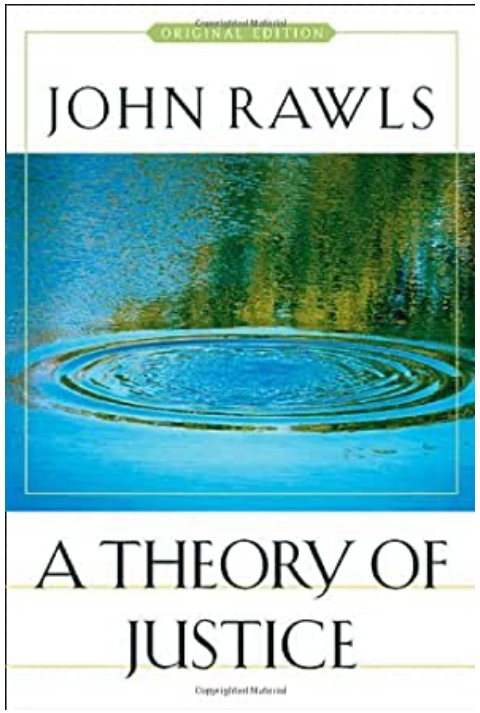
η = ∞: total inequality aversion

John Rawls's *Theory of Justice* (1971)

Extreme inequality aversion



John Rawls (1921-2002)



(1971, rev. 1999)

- ToJ = the seminal statement of post-war liberal egalitarianism.
- ~2.500 books & articles written on ToJ 10y after its publication.
- Lays out an account of a just society with a strong aversion against unequal distributions of primary resources.
- One of the most influential philosophical justifications of redistributive policies to date.

Rawls' Two Principles of Justice

What is a just society? How much economic inequality is tolerable?

1. Each person is to have an equal right to the most extensive total system of **equal basic liberties** compatible with a similar system of liberty for all.
2. Social and economic inequalities are to be arranged so that they are both:
   2.1. attached to offices and positions open to all under conditions of fair **equality of opportunity**, and
   2.2. to the greatest benefit of the **least advantaged**.

The "Difference Principle"

What Society Would You Choose to Be Reborn In?

Rawls' bet: under the veil, rational parties would choose the basic structure (think: constitution) in which your hypothetical worst case is better than your hypothetical worst case in any other alternative economic structure.

- **Maximin**: "maximise the minimal (worst possible) outcome"
- Why not choose average utility?

Society	Red	Blue	Green	Average Income
A	100	100	100	100
B	150	100	50	100
C	600	600	99	433
D	1096	102	101	433

Policy Implications of Inequality Aversion

Four Welfare State Regimes

Welfare regime	Ideological goal	Main provider of welfare	Workforce emancipation from the labour market	Degree of stratification	Allocation of social rights	Involvement of the state in providing welfare	Historical political coalitions	Archetypical examples
Liberal	To promote individualism and **traditional work ethic** through the labour market.	The labour market (individual)	**Lowest** – welfare benefits are directly tied to participation in the labour market.	**Moderate**, with class dualism between the poor state welfare recipients, and the majority who secure insurance via the private sector.	Market differentiated – based on individual needs and means-tests.	**Residual**, providing only a "safety net" at the margin.	Right-wing dominant	United States, Canada, Australia.
Corporatist	To promote traditional values such as **familyhood**, preserve class differentials and legitimize the central authority.	Kinship (family)	**Low** – welfare benefits are tied to employment, although to a lesser extent than in the liberal model.	**Highest**, due to prioritizing the preservation of class politics.	Based on class strata and tied to contributions.	**Central** – private welfare providers are crowded out.	Conservative, traditional, Catholic influence	Austria, France, Germany, Italy.
Social-democratic	To promote **equality** of rights and benefits enjoyed by all members of society.	State	**High**, due to the presence of labour unions and welfare benefits being largely independent of employment, but regime still reliant on production through passive labour market policies.	**Lowest**, with the focus being equality of rights, benefits and status.	Universal, egalitarian access to rights.	**Main provider**	Left-wing dominant	Sweden, Norway, Denmark.

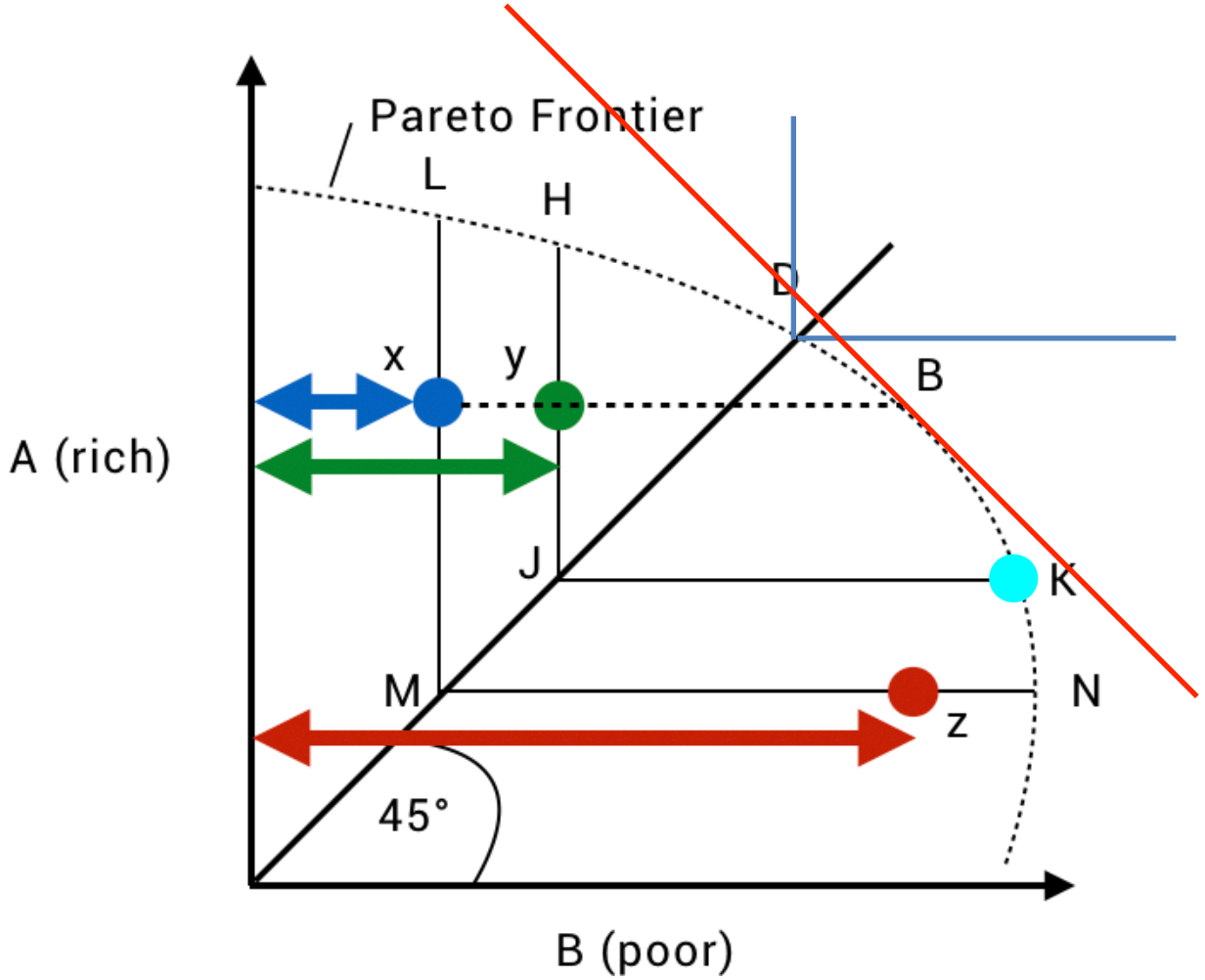
Adjusted from Lukas Latauskas & Eriks Sniedze (2020). *Typology of the Latvian Welfare State*. SSE Riga BSc Thesis.

Bentham, Pareto, and Rawls Compared

x = starting point

What to say about changes from:

- x to H or D?
- x to y?
- x to J?
- x to K?
- x to z?



Jointly Increasing Efficiency and Fairness?

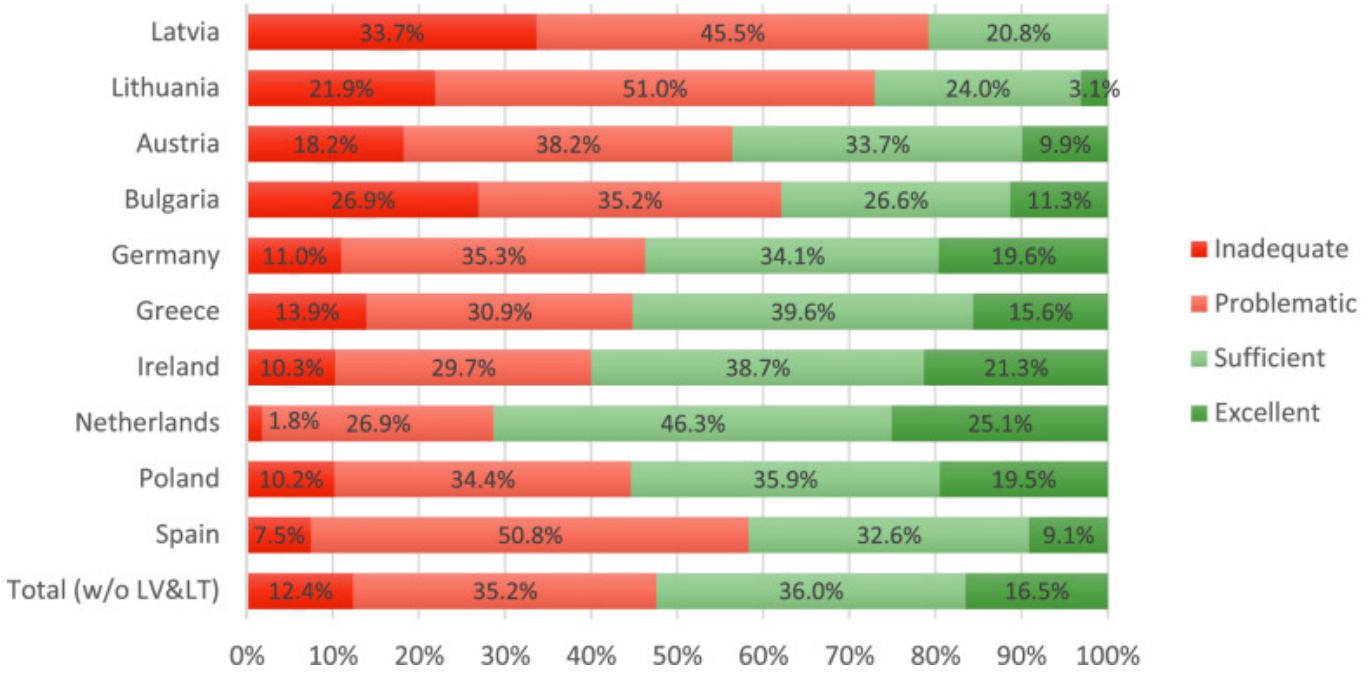
Consider the efficiency and fairness impact of the following policies:

- Increasing the effectiveness of antitrust laws.
- Tightening regulations on management compensation in the financial sector.
- Restricting (or increasing the transparency of) lobbying activities within legislatures.
- Reforming tax policies that tax capital gains at a low rate.
- Improving educational performance in public schools.
- Improving health literacy.

Example: Health Literacy

"Cognitive and social skills which determine the motivation and ability of individuals to gain access to, understand and use information in ways which promote and maintain good health."

— World Health Organization



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Example: Health Literacy

Relationship with health costs and health equity

Efficiency cost of low health literacy:

- Low health literacy is empirically associated with:
  - more visits to the general practitioner.
  - increased rates of hospitalisations (in-patient care).
  - longer stays in hospitals.
  - higher rates of emergency service use.
  - under-utilisation of preventive healthcare services.

Equity cost of low health literacy:

- Health equity: “the attainment of the highest level of health for all people” (WHO).
- Provisional evidence (from the US) showing that health literacy mediates socio-economic determinants of health, particularly education, age, income, and race/ethnicity.
- Hypothesis: health literacy critically influences how social disadvantages translate into disparate health outcomes.

When Governments Fail to Promote the Social Good (Efficiency, Fairness…)

Important Causes of Government Failure

1. Preference aggregation problems: when social welfare functions fail to return a coherent social preference (reveal “the will of the people”).
2. Rent seeking: when private actors secure non-market rents through lobbying.
3. Problems of district-based representation: when policymakers mistake social benefits for benefits limited to their districts.
4. Policy myopia: when policymakers’ calculations of social benefits depend on electoral time horizons.
5. Bureaucratic failure: when there is misalignment between the interests of the government and of bureaucrats.
6. Fiscal externalities: when constituencies engage in a ruinous competition for low taxes.

Preference Aggregation Problems

Aggregation Procedures

"How do we get to know the will of society?"

Remember: the point of a social welfare function (SWF) is to take as inputs all of the individual preferences and return as output a social preference! It can be conceptualised as:

- A mathematical function.
- A method for aggregating individual preferences.

Let A bet the set of policy alternatives for a given issue. Individuals have preferences over all alternatives in the set A. So for instance in tax policy: **A = { No tax, low tax, medium tax, high tax }**

A social preference is simply a rank ordering of all the alternatives in the set A.

What are some aggregation procedures?

Example 1: Pairwise Majority Voting

A society with three members — Beth (B), Charles (C), and Dana (D) — is considering tax policy. There are four possible taxes: None, Low, Medium, and High.

Hence: the set of alternatives is A = {N, L, M, H}.

The three members have the following preferences:

$H >_B N >_B L >_B M$

$M >_C H >_C L >_C N$

$M >_D H >_D N >_D L.$

According to pairwise majority voting, the social preference will be:

Pairwise Majority Rule

For any two alternatives, x, y ∈ A, society prefers x to y if a majority of individuals in society prefers x to y, society prefers y to x if a majority of individuals prefers y to x, and society is indifferent between x and y if the same number of people prefers x to y as y to x.

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According to pairwise majority voting, the social preference will be:

$M \succ_{maj} H \succ_{maj} N \succ_{maj} L$ **M is the Condorcet winner!**

Borda Count Aggregation

Consider the same society with the same set of policy alternatives.

The three members have the following preferences:

$H >_B N >_B L >_B M$

$M >_C H >_C L >_C N$

$M >_D H >_D N >_D L.$

According to the Borda Count, the social preference will be:

Borda Count:

If there are *k* alternatives, each person's first choice gets k−1 points, each person's second choice gets k−2 points, and so on, with each person's last choice getting 0 points

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According to the Borda Count, the social preference will be:

$H >_{bor} M >_{bor} N >_{bor} L$

Scores:

H: 3 + 2 + 2 = 7

M: 0 + 3 + 3 = 6

L: 1 + 1 + 0 = 2

N: 2 + 0 + 1 = 3

Borda Count:

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What Makes for a “Good” Aggregation Procedure?

We want aggregation procedures to fulfil the following basic criteria:

1. **Transitivity** of social preferences: if some aggregation procedure yields intransitive social preferences, what is then the “will of the people” (or the “public interest”?)

   But consider pairwise majority voting given this profile:

   $x >_1 y >_1 z$

   $y >_2 z >_2 x$

   $z >_3 x >_3 y$

   Social preference is intransitive!! (Condorcet Paradox)

   $\mathbf{x >_{maj} y >_{maj} z >_{maj} x.}$

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   $x \succ_{maj} y \succ_{maj} z \succ_{maj} x$.

Based on: Ethan Bueno de Mesquita. 2016. *Political Economy for Public Policy*. Princeton University Press. p. 53

The Money Pump

- Paul: Apples > Bananas, Bananas > Oranges, and Oranges > Apples.
- Paul values each of these preferences at $1.10.
- Suppose Paul has an orange and $10.
- Sally is a fruit seller. One day, she offers to trade Paul one of her bananas in exchange for Paul's orange plus a dollar. Paul accepts.
- The next day, Sally offers Paul an Apple in exchange for Paul's banana and a dollar. Paul accepts.
- What will be the endgame?

What Makes for a “Good” Aggregation Procedure?

We want aggregation procedures to fulfil the following basic criteria:

2. **Unanimity** of social preferences: an aggregation procedure satisfies unanimity if, whenever everyone prefers some alternative x to another alternative y, the aggregation procedure says that society prefers x to y as well.

   But consider sequential (tournament style) run-off voting:

   $A >_1 B >_1 D >_1 C$ Vote 1: A vs. B (A wins)

   $C >_2 A >_2 B >_2 D$ Vote 2: A vs. C (C wins)

   $B >_3 D >_3 C >_3 A$ Vote 3: C vs. D (D wins)

   BUT everyone prefers B > D!

What Makes for a "Good" Aggregation Procedure?

We want aggregation procedures to fulfil the following basic criteria:

3. **Independence of Irrelevant Alternatives (IIA)**: the social preference between two alternatives, x and y, should depend only on people's preferences over x and y. But it should not depend on what people think about some irrelevant alternative, z.

   Now, suppose voter 1 moves B from second place to last:

$A >_1 \mathbf{B} >_1 C >_1 D$	$A >_1 C >_1 D >_1 \mathbf{B}$
$B >_2 A >_2 C >_2 D$	$B >_2 A >_2 C >_2 D$
$B >_3 A >_3 C >_3 D$	$B >_3 A >_3 C >_3 D$
(B = 8, A = 7, C = 3, D = 0)	(**A = 7**, B = 6, C = 4, D = 1)

But: Nothing has changed regarding the individual rankings of A and B!! The only difference? For V1, C and D are now "between" A and B.

Kenneth Arrow's (Im-)Possibility Theorem

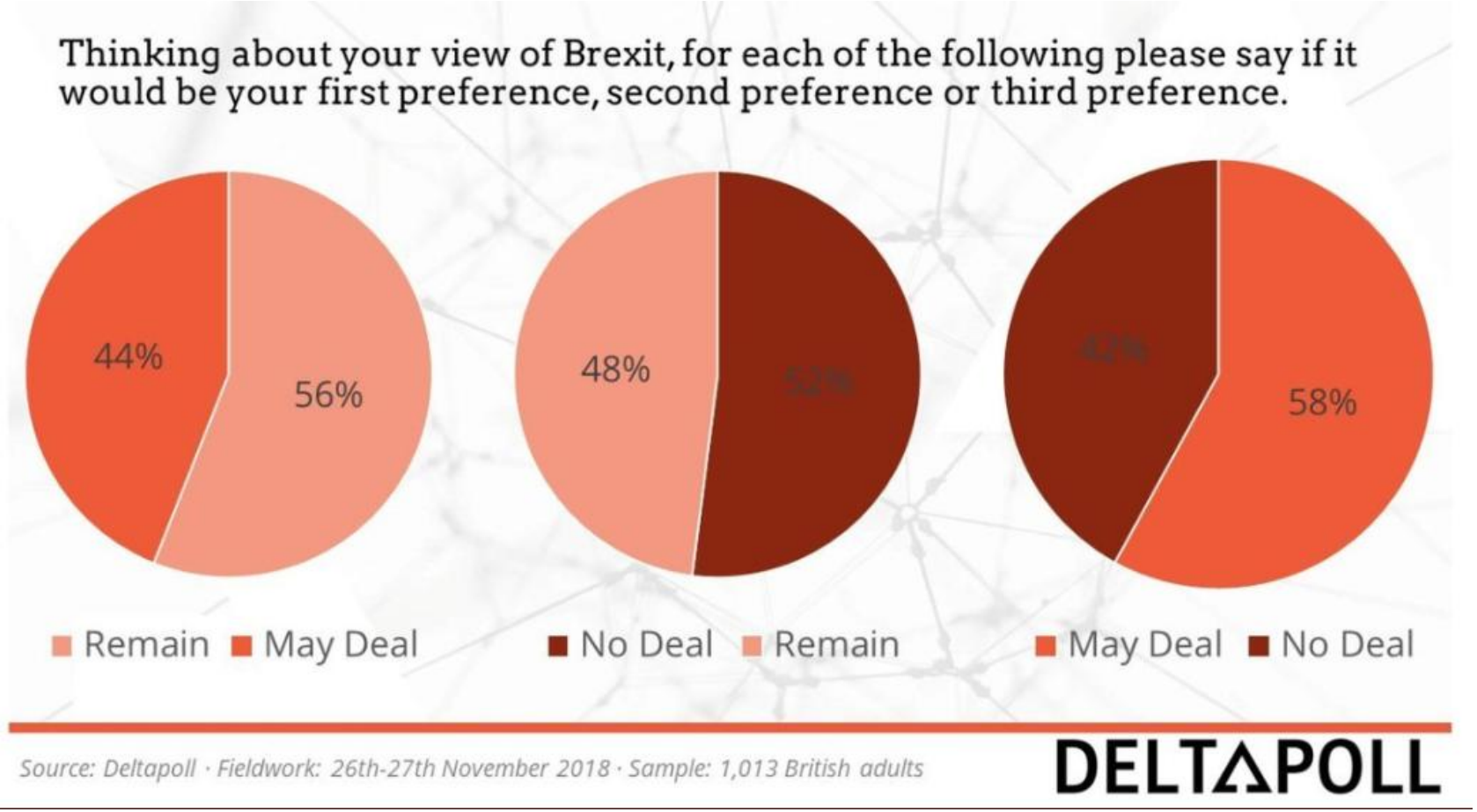
Which aggregation procedure guarantees a coherent outcome?

If there are at least two people considering at least three policy alternatives, then no **non-dictatorial** social welfare function can simultaneously satisfy transitivity, unanimity, and IIA.

- Kenneth J. Arrow. 1950. "A Difficulty in the Concept of Social Welfare." *Journal of Political Economy*, 58(4): 328-346.
- Our capacity to choose rationally can be lost when individual preferences are aggregated into social preferences.
- Bad news for democracy and majority voting! Democracy can and will produce cycles!
  - Policymakers may never know the "will of the people" in any coherent way.
  - See William Riker and the "social choice" objection to democracy.
  - The (University of) Rochester school of economics.

Example: Brexit Polls In 2018-2019

Arrow's Theorem in the real world



1) R>D
2) N>R
3) D>N

Rent-Seeking

Rent-Seeking

The pursuit of economic benefits through influencing the policy process that would not be achievable through competitive markets.

"Particular industries and occupations obtain from the state a variety of economic privileges which are injurious to the vast majority of the population. Farm subsidies, oil import quotas, tariffs, and occupational licensing are examples."

— Stigler, George J. 1972. "Economic Competition and Political Competition." *Public Choice* 13(1): p. 100.

Concentrated Benefits and Diffuse Costs

Table 2.2 Wilson's typology.

Costs	*Benefits Concentrated*	*Diffuse*
Concentrated	Interest group politics ('zero-sum game')	Entrepreneurial politics
Diffuse	Clientelistic politics	Majoritarian politics

Sources: Based on Wilson (1989) and Knill and Tosun (2011).

Original statement: James Q. Wilson. 1973. *Political Organizations*. Basic Books.

Example: Price Floors

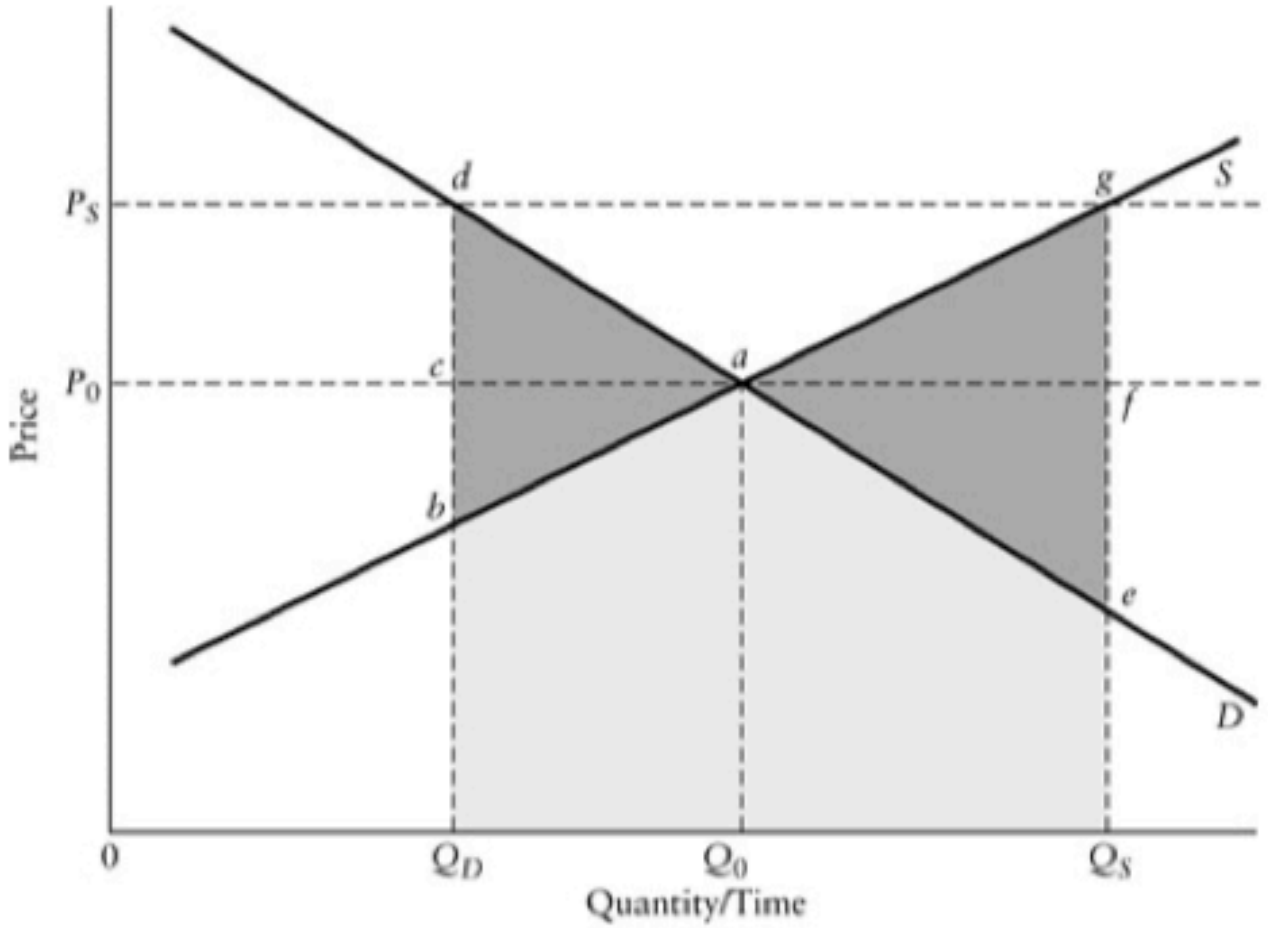
With a price floor at P_S, producers get a rent o P_SP_0cd.

Welfare impact?

1. -abd in the case of imposed production quota.
2. -aeg in the case of government purchase with distribution.
3. -$Q_D$dagQ_S in the case of government purchase with destruction.

"Transitional Gains Trap:"

Undoing price floors can have devastating consequences for current [land] owners.



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Example: Price Ceilings

Best case:

With a price ceiling at P_C, consumers get a rent of P_0P_Cbc.

Welfare impact? -bad

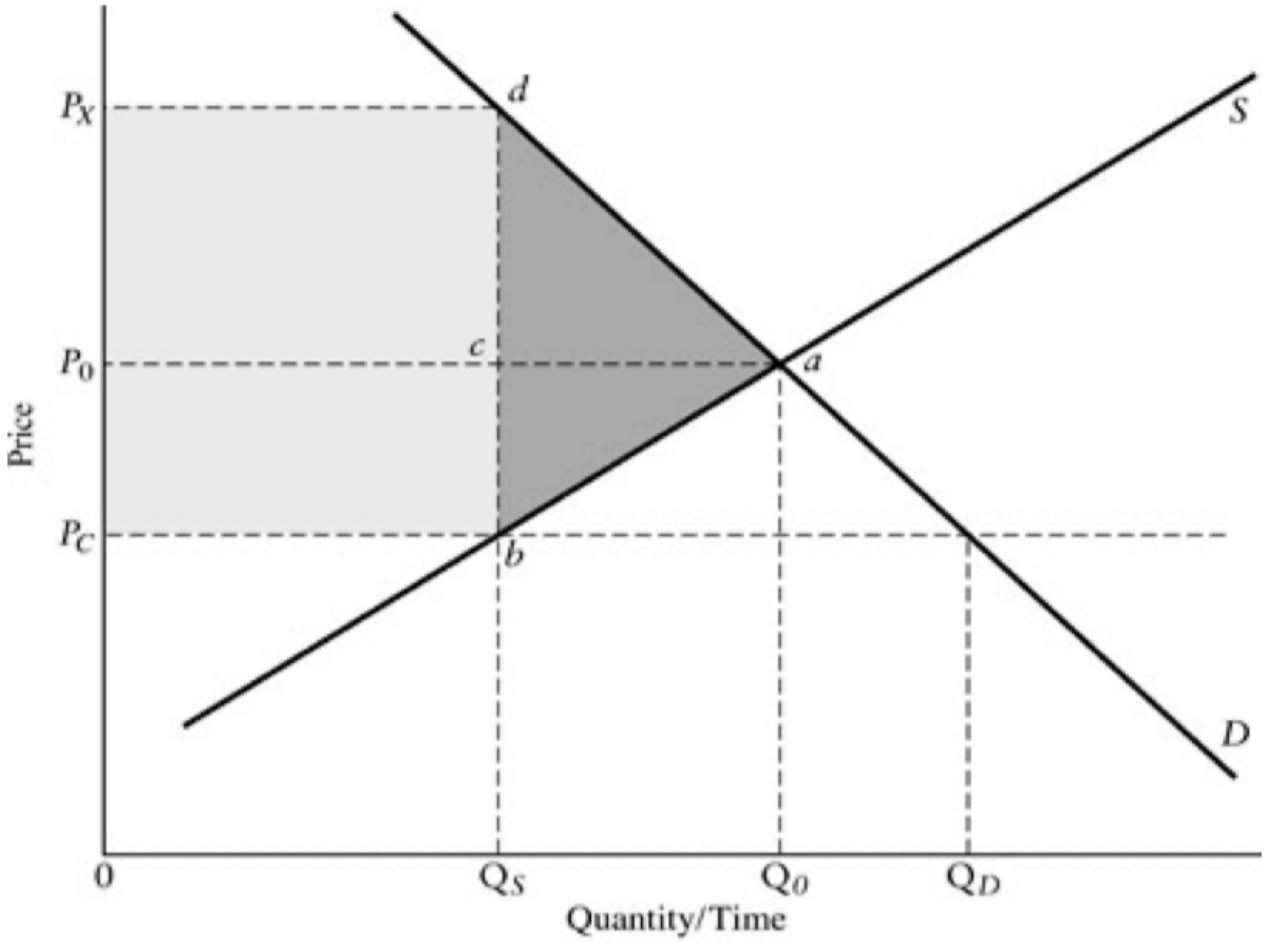
Worst case:

Consumers value Q_S at P_X. They will "pay" up to P_X-P_C in non-monetary costs for getting shares of the scarce good!

In some cases, consumers dissipate their total surplus P_XP_Cbd!

Welfare impact? -P_XP_Cbad

Fairness impact?



Thank You